Supplementary Material

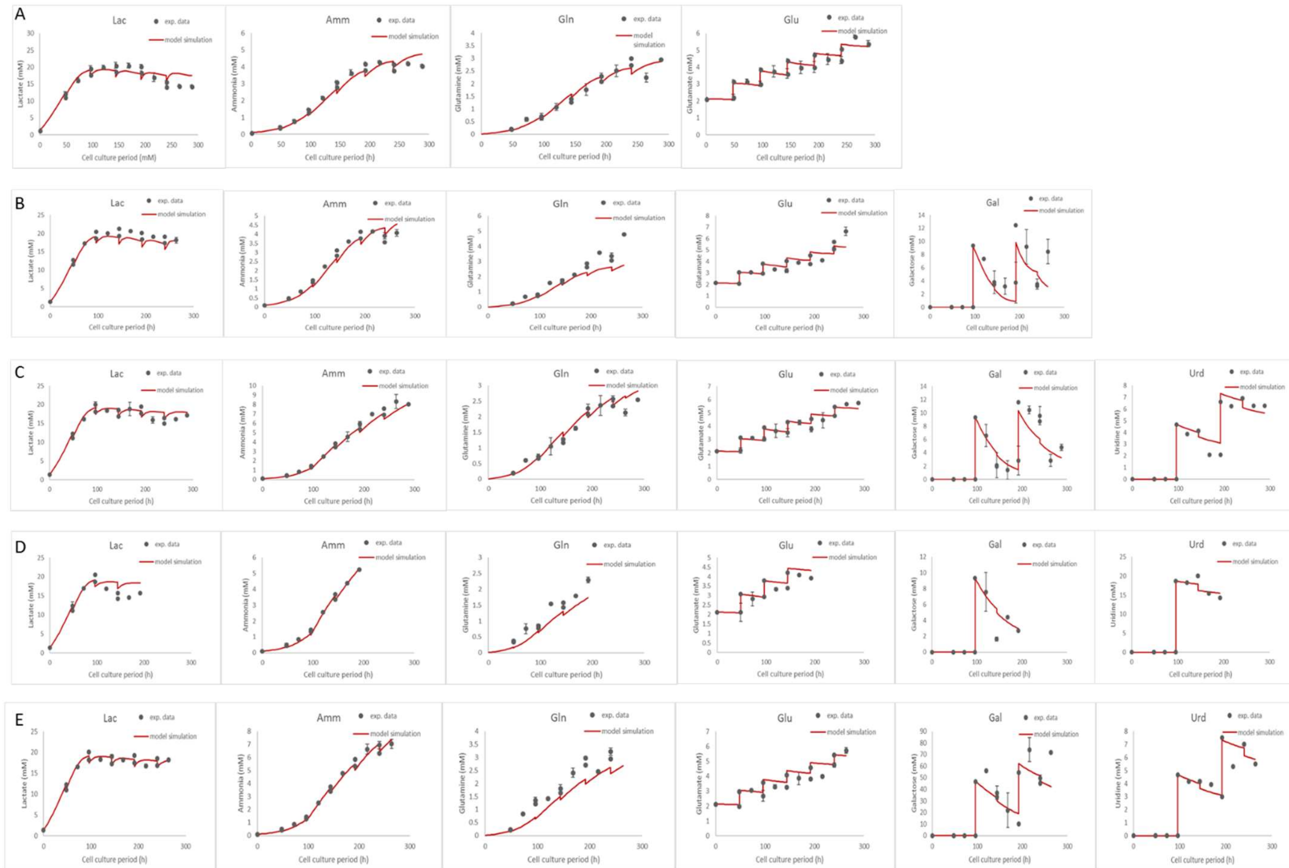

Figure S1. Remaining graphs of cell metabolism model simulation for all experiments used for model construction and parameter estimation. Row A presents the fitting of the model to the control experiment, row B the fitting of the model to the 10G experiment, row C the fitting of the model to the 10G5U experiment, row D the fitting of the model to the 10G20U experiment and row E the fitting of the model to the 50G5U experiment.

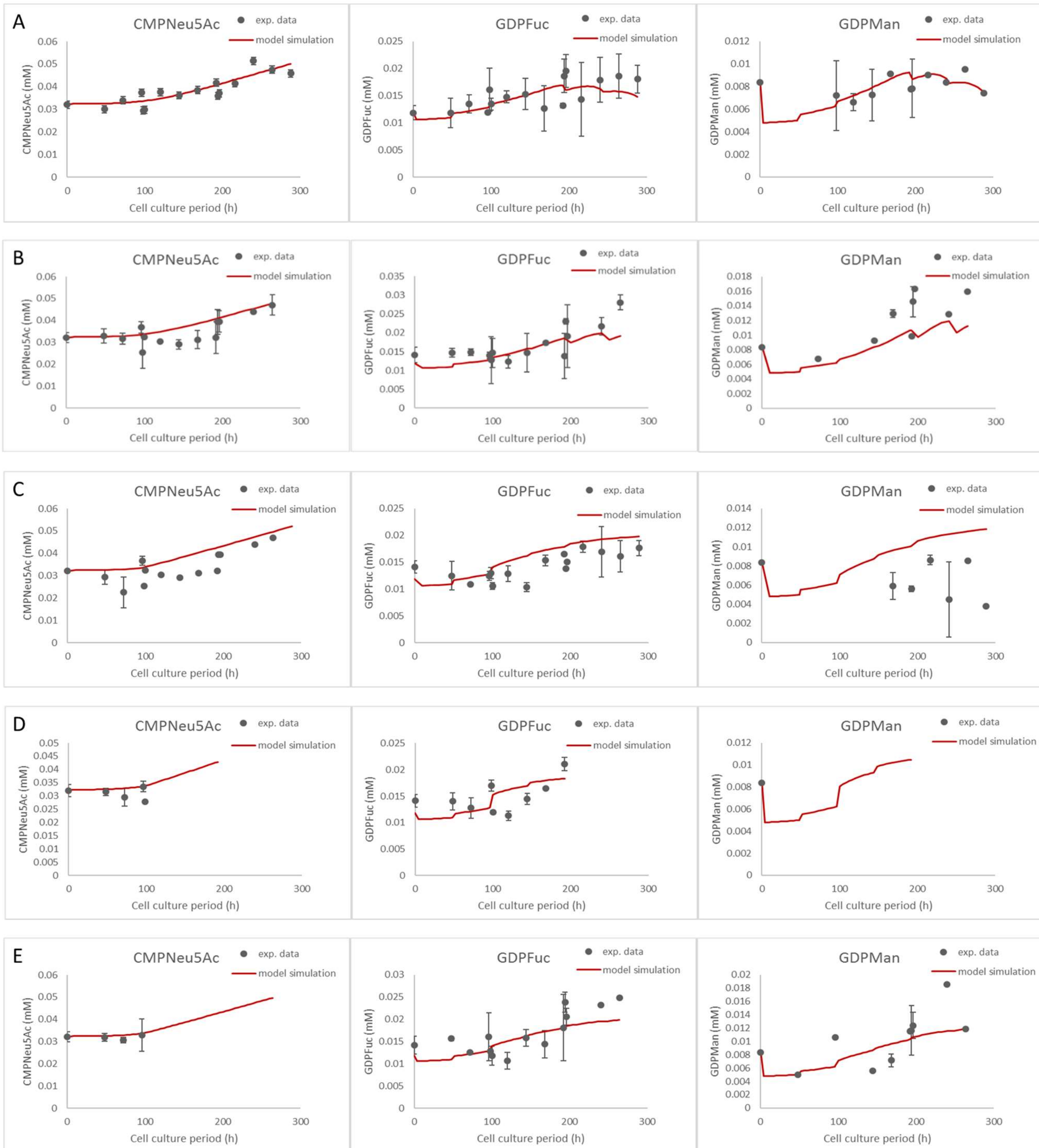

Figure S2. Remaining graphs of the NSD synthesis model simulation for all experiments used for model construction and parameter estimation. Row A presents the fitting of the model to the control experiment, row B the fitting of the model to the 10G experiment, row C the fitting of the model to the 10G5U experiment, row D the fitting of the model to the 10G20U experiment and row E the fitting of the model to the 50G5U experiment.

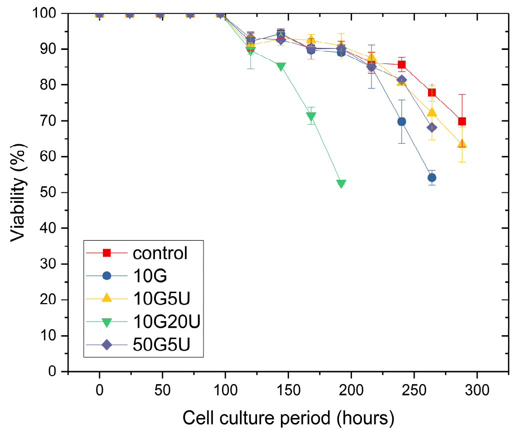

Figure S3. Cell viability for all the experiments used for model construction and parameter estimation.

Table S1. Estimated parameters for the cell metabolism and NSD synthesis model

Parameter	Value	Unit	95% Confidence Interval
cell metabolism			
$Y_{X_{Glc}}$	1.01×10^{9}	cell·mmol^{-1}	1.50×10^{8}
$Y_{X_{Lac}}$	5.46×10^{7}	cell·mmol^{-1}	7.60×10^{6}
$Y_{X_{Gln}}$	4.64×10^{9}	cell·mmol^{-1}	2.05×10^{8}
$Y_{X_{Glu}}$	1.46×10^{10}	cell·mmol^{-1}	3.29×10^{9}
$Y_{X_{Asn}}$	7.68×10^{8}	cell·mmol^{-1}	7.23×10^{6}
$Y_{X_{Amm}}$	2.36×10^{9}	cell·mmol^{-1}	2.15×10^{7}
$Y_{X_{Gal}}$	1.38×10^{8}	cell·mmol^{-1}	4.09×10^{6}
$Y_{X_{Urd}}$	1.61×10^{9}	cell·mmol^{-1}	2.39×10^{7}
$Y_{X_{Asp}}$	3.59×10^{9}	cell·mmol^{-1}	fixed [1]
$Y_{Gln/Amm}$	0.10	mmol·mmol^{-1}	2.24×10^{-2}
$Y_{Lac/Glc}$	1.56	mmol·mmol^{-1}	fixed [2]
$Y_{Asn/Asp}$	0.10	mmol·mmol^{-1}	0.03
$Y_{Asp/Asn}$	0.13	mmol·mmol^{-1}	fixed [1]
$Y_{Amm/Urd}$	2	mmol·mmol^{-1}	0.03
m_{Glc}	3.43×10^{-11}	mmol·cell^{-1}·h^{-1}	1.50×10^{-12}
m_{lac}	1.87×10^{-10}	mmol·cell^{-1}·h^{-1}	4.65×10^{-11}
Kc_{Gal}	5.27	mM	0.31
f_{Gal}	0.35	N/A	0.03
Lac_{max1}	21.20	mM	0.73
Lac_{max2}	16	mM	0.50
K_{Gal}	18.23	mM	0.74
K_{Urd}	7	mM	0.26
NSD synthesis			
f_{Gln}	2.00×10^{-2}	N/A	3.30×10^{-4}
$K_{M1_{Gln}}$	0.42	mM	6.23×10^{-3}
$K_{M1_{sink}}$	4.00×10^{-2}	mM	1.98×10^{-3}
$KI_{1_{sink}}$	1.21×10^{-4}	mM	2.18×10^{-6}
$K_{M2_{Glc}}$	78.12	mM	6.719
$K_{M2b_{UDPGal}}$	2.48×10^{-2}	mM	3.22×10^{-3}
KI_{2A}	1.05×10^{-6}	mM	1.54×10^{-7}
KI_{2B}	92.11	mM	9.82×10^{7}
KI_{2C}	1.33×10^{-2}	mM	1.28
KI_{2D}	2.66×10^{-6}	mM	1.23×10^{-7}
$K_{M3_{Glc}}$	50	mM	4.05
$K_{M4_{UDPGlcNAc}}$	2.31	mM	0.30
$K_{M5_{UDPGlcNAc}}$	2.70×10^{-2}	mM	8.21×10^{-3}
KI_{5}	1000	mM	8.92×10^{6}
$K_{M6_{UDPGlc}}$	1.6×10^{-2}	mM	3.22×10^{-4}
KI_{6A}	1.10×10^{-7}	mM	6.73×10^{-9}
KI_{6B}	4.57	mM	2.81×10^{5}
KI_{6C}	4.84×10^{-6}	mM	5.08×10^{-7}
$K_{6_{sink}}$	0.13	mM	1.88×10^{-3}
$KI_{6_{sink}}$	10	mM	3.83×10^{-3}

$K_{regulator}$	1.00×10^{-5}	mM	fixed
$K_{M7_{GDPMan}}$	0.99	mM	0.10
KI_7	1.64×10^{-2}	mM	3.57×10^{-3}
$K_{M7_{sink}}$	8.88	mM	0.65
$V_{max,1_{Urd}}$	0.15	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	6.02×10^{-3}
$V_{max,2_{Urd}}$	4.52×10^{-2}	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	4.23×10^{-4}
$V_{max,4_{Urd}}$	1.27×10^{-2}	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	2.70×10^{-3}
$V_{max,6_{Urd}}$	5.34	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	1.10×10^{-2}
$V_{max7_{sink}}$	10.94	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	0.81
$K_{M,1_{Urd}}$	6.08	mM	0.51
$K_{M,2_{Urd}}$	13.63	mM	0.69
$K_{M,4_{Urd}}$	6.25	mM	2.62
$K_{M,6_{Urd}}$	0.44	mM	1.91×10^{-3}
$V_{max1_{sink}}$	25.49	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	0.46
$V_{max6_{sink}}$	7.30	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	0.04
$V_{max6_{Gal}}$	40.90	$mmol_{NSD}\cdot L_{cell}^{-1}\cdot h^{-1}$	9.86
$K_{M6_{Gal}}$	0.60	mM	0.14
KI_{6D}	1.00×10^{-2}	mM	3.75×10^{-4}
KI_{6E}	99.63	mM	525.2
KI_{6F}	9.11×10^{-4}	mM	8.74×10^{-6}
$K_{TP_{UDPGlc}}$	0.90	mM	4.47×10^{-2}
$K_{TP_{UDPGlcNAc}}$	5.05	mM	2.31
$K_{TP_{UDPGal}}$	7.15	mM	0.33
$K_{TP_{UDPGalNAc}}$	11.06	mM	9.12
$K_{TP_{GDPMan}}$	0.13	mM	3.42×10^{-2}
$K_{TP_{GDPFuc}}$	0.10	mM	0.75
$K_{TP_{CMPNeu5Ac}}$	503.21	mM	824
V_{cell}	1.12×10^{-12}	L	fixed
V_{Golgi}	2.50×10^{-14}	L	fixed
MW_{mAb}	165.17×10^{3}	$g_{mAb}\cdot mol_{mAb}^{-1}$	fixed
$N_{HCP/Lipids_{UDPGlc}}$	1.56×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{UDPGlcNAc}}$	1.25×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{UDPGal}}$	2.29×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{UDPGalNAc}}$	1.25×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{GDPMan}}$	3.54×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{GDPFuc}}$	1.40×10^{-13}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{HCP/Lipids_{CMPNeu5Ac}}$	1.85×10^{-12}	$mmol_{NSD}\cdot cell^{-1}$	fixed [3]
$N_{precursor_{UDPGlc}}$	40.39×10^{-6}	$mmol_{NSD}\cdot mg_{mAb}^{-1}$	fixed [3]
$N_{precursor_{GDPMan}}$	121.2×10^{-6}	$mmol_{NSD}\cdot mg_{mAb}^{-1}$	fixed [3]
$N_{precursor_{UDPGlcNAc}}$	26.67×10^{-6}	$mmol_{NSD}\cdot mg_{mAb}^{-1}$	fixed [3]

Table S2. Estimated enzyme distribution parameters and dissociation constants

Estimated parameter	**Value**	**Units**	**95% Confidence Interval**[a]
$ManI_{max}$	1.034	μM	N/A
ω_{ManI}	1.160	-	N/A
$ManII_{max}$	0.854	μM	N/A
ωMan_{II}	0.953	-	N/A
$GnTI_{max}$	1.870	μM	N/A
$\omega_{Gn}TI$	0.300	-	N/A
$GnTII_{max}$	1.566	μM	N/A
ω_{GnTII}	0.912	-	N/A
$GalT_{max}$	0.464	μM	N/A
ω_{GalT}	0.109	-	N/A
$FucT_{max}$	0.349	μM	N/A
ω_{FucT}	0.470	-	N/A
Kdi_{FucTB}	11.2	μM	0.404
$Kdi_{GalTa1A}$	96.4	μM	2.653
$Kdi_{GalTa1B}$	345.8	μM	37.75
Kdi_{GnTI}	234.4	μM	7.15
$Kdk_{GalTa1A}$	1805.0	μM	213.2
$Kdk_{GalTa1B}$	1827.1	μM	1591

[a] The values have been obtained via optimisation without using experimental data where 'N/A' appears; therefore, there are no 95% confidence intervals to report.

Table S3. Specific production rate of galactosylated mAb

	$q_{gal,mAb}$ (pg·cell^{-1}·day^{-1})				
time (day)	**control**	**10G**	**10G5U**	**10G20U**	**50G5U**
7	5.59	6.95	7.65	7.27	8.30
9	4.81	6.32	6.34	-	6.13
11	4.26	4.77	5.37	-	5.29
12	4.22	-	5.37	-	-

References

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